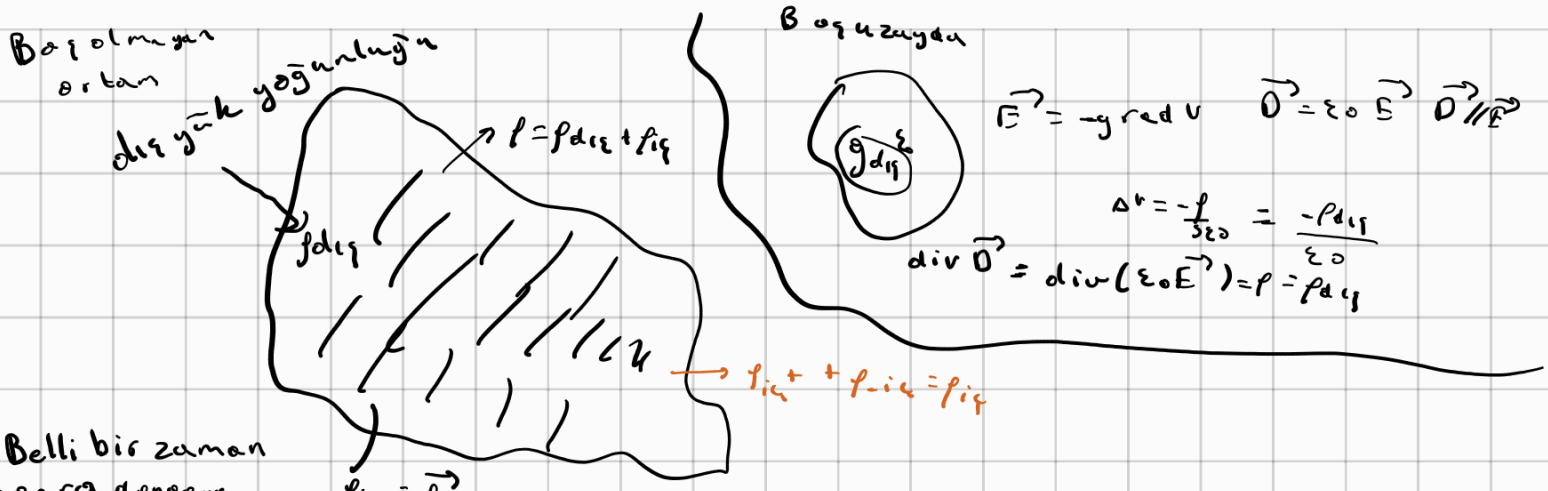


$\vec{E} = -\text{grad}\, V \qquad \vec{D} = \varepsilon_0 \vec{E} \qquad \vec{D} \parallel \vec{E}$

$\Delta V = -\frac{\rho}{3\varepsilon_0} = \frac{-\rho_{dış}}{\varepsilon_0}$

$\text{div}\, \vec{D} = \text{div}(\varepsilon_0 \vec{E}) = \rho = \rho_{dış}$

($\vec{P}$: Ortamın fiziksel yapısından dolayı dış yük tarafından yaratılan $\vec{E}$'nin etkisiyle oluşan yük yoğunluğu)

$\rho = \rho_{dış} + \bar{\rho}$ artık ortamı boş gibi düşünebiliriz



$\text{div}(\varepsilon_0 \vec{E}) = \rho = \rho_{dış} + \bar{\rho}$

$\bar{\rho} = -\text{div}(\varepsilon_0 \chi(\vec{E})\, \vec{E})$ (χ: kapa katsayısı, ortamın duyarlılığıdır)

div lineerdir

$\text{div}(\varepsilon_0 \vec{E}) - \bar{\rho} = \rho_{dış}$

$\text{div}(\varepsilon_0 \vec{E}) + \text{div}(\varepsilon_0 \chi(\vec{E})\, \vec{E}) = \rho_{dış}$

$\text{div}(\varepsilon_0 (1 + \chi(\vec{E}))\, \vec{E}) = \rho_{dış}$ ($\varepsilon_r > 1$)

İçinde bir takım dipollerin oluştuğu ortama dielektrik ortam denir. Böyle bir ortamda $\vec{P} = \varepsilon_0 \chi(\vec{E}) \cdot \vec{E}$, $\vec{E}$ ile aynı yönde

dolayısıyla da $\vec{P} \parallel \vec{E} \Rightarrow \chi(\vec{E}) > 0$

aynı yönde ve paraleldir.

$\text{div}(\underbrace{\varepsilon_0 \varepsilon_r}_{\varepsilon} \cdot \vec{E}) = \rho_{dış} \Rightarrow \text{div}(\varepsilon \cdot \vec{E}) = \rho_{dış}$

$\vec{D} = \varepsilon \vec{E}$

$\text{div}\, \vec{D} = \rho_{dış}$

ortam basitse kapa elektrik alana bağımlı değil sabittir.

Basit ortamda $\varepsilon = \varepsilon_0 \cdot \varepsilon_r \Rightarrow \varepsilon > \varepsilon_0$ $\qquad \varepsilon_r = \frac{\varepsilon}{\varepsilon_0} > 1$ bağıl dielektrik sabiti

Boşluk: $\text{div}\, \vec{D} = \rho_{dış}$ ($\vec{D} = \varepsilon_0 \vec{E}$)

$\text{div}\, \vec{E} = -\rho_{dış}/\varepsilon_0$ $\qquad \text{div}\, \vec{D} = \rho_{dış}$

$\Delta V = -\rho_{dış}/\varepsilon_0$ $\xrightarrow{\varepsilon \vec{E}}$ $V = \frac{q}{4\pi\varepsilon_0 r}$

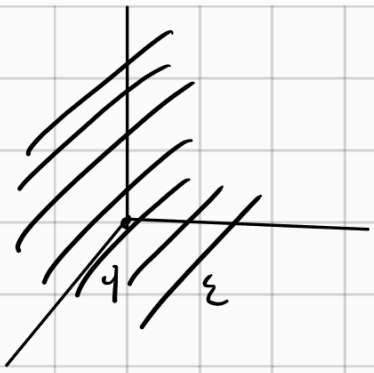
$\text{rot}\, \vec{E} = 0$

$\text{div}\, \vec{D} = \rho_{dış}$ $\qquad \vec{F} = q \vec{E}$

$\Delta V = -\frac{\rho_{dış}}{\varepsilon}$ ortam basitse

$\Rightarrow \text{div}(\varepsilon \vec{E}) = \rho_{dış}$

$\text{div}\, \vec{E} = \frac{\rho_{dış}}{\varepsilon}$



ε

$q = 1\,C\,ise$

$$V = \frac{1}{4\pi\varepsilon \cdot r}$$

$$V = \frac{1}{4\pi\varepsilon} \int \frac{\rho}{r_1} \, d\vartheta$$

$$\vec{E} = -\,grad\,V \qquad \vec{D} = \varepsilon \vec{E} \qquad rot\,\vec{E} = 0$$



$$\vec{n} = \frac{d\vec{s}}{ds}$$

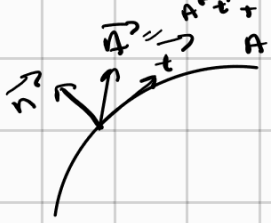
$$rot\,\vec{E} = 0 \Rightarrow \{rot\,\vec{E}\} + \vec{n} \times (\vec{E}_2 - \vec{E}_1)\,l_s \; S(s) = \{0\} + 0 \cdot S(s)$$

$$\left\{ \begin{matrix} rot\,\vec{E}_1 : v_1 \\ rot\,\vec{E}_2 : v_2 \end{matrix} \right\} + \vec{n} \times (\vec{E}_2 l_s - \vec{E}_1 l_s)\, S(s)$$

$$= \left\{ \begin{matrix} 0\; v_1 \\ 0\; v_2 \end{matrix} \right\} + 0\, S(s)$$

$$\vec{n} \times (\vec{E}_2 l_s - \vec{E}_1 l_s) = 0$$

$$\Rightarrow \vec{n} \times \vec{E}_2 l_s = \vec{n} \times \vec{E}_1 l_s \Rightarrow \vec{E}_1{}^t l_s = \vec{E}_2{}^t l_s$$



$$\vec{n} \times \vec{A} = \vec{n} \times (A^t \vec{t} + A^n \vec{n}) = A^t (\vec{n} \times \vec{t}) + A^n \underbrace{\vec{n} \times \vec{n}}_{0}$$

$$\vec{n} \cdot \vec{A} = \vec{n} (A^t \vec{t} + A^n \vec{n})^t = A^t \vec{n} \cdot \vec{t} + A^n \underbrace{\vec{n} \cdot \vec{n}}_{1} = A^n$$

$$div\,\vec{D} = \rho$$

$$\hookrightarrow \begin{cases} div\,\vec{D}_1 : v_1 \\ div\,\vec{D}_2 : v_2 \end{cases} + \vec{n}(\vec{D}_2 - \vec{D}_1)\Big|_S = \begin{cases} g_1 ; v_1 \\ g_2 ; v_2 \end{cases} + \rho_S S(s)$$

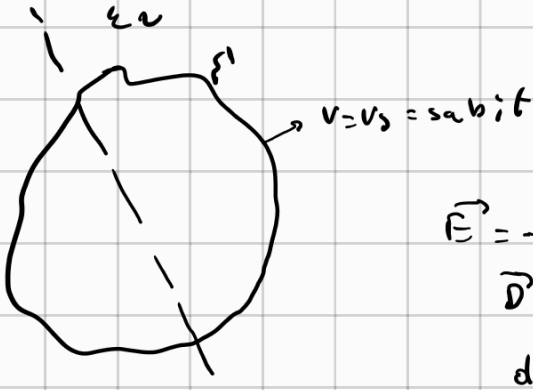
$$div\,\vec{D}_1 = \rho_1 \qquad \vec{n}(\vec{D}_2 - \vec{D}_1)\,|_S = \rho_S$$
$$div\,\vec{D}_2 = \rho_2 \qquad \Rightarrow \vec{n}\,\vec{D}_2|_S - \vec{n}\,\vec{D}_1|_S = \rho_S$$

$$D_2^{\,n}|_S - D_1^{\,n}|_S = \rho_S = -\varepsilon_2 \frac{\partial v_2}{\partial n}\Big|_S + \varepsilon_1 \frac{\partial v_1}{\partial n}\Big|_S = \rho_S$$
$$\Rightarrow +\varepsilon_2 \frac{\partial v_2}{\partial n}\Big|_S - \varepsilon_1 \frac{\partial v_1}{\partial n}\Big|_S = -\rho_S$$

$$\vec{E} = -grad\,v = \begin{cases} -grad\,v_1 : v_1 e^{i\varphi} \\ -grad\,v_2 : v_2 e^{dig} \end{cases} + \overbrace{\vec{n}(v_2 - v_1)|_S}^{0}\, S(s)$$

$$\boxed{v_1|_S = v_2|_S}$$

$$\Delta v_1 = -\frac{\rho_1}{\varepsilon_1} \qquad v_1' d\varepsilon$$
$$\Delta v_2 = -\frac{\rho_2}{\varepsilon_2} \qquad v_2' d\varepsilon$$



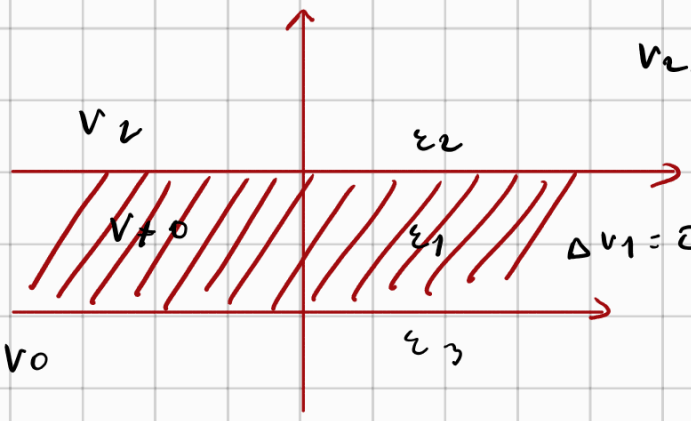
$$\vec{E} = -grad\,v = -grad\,v_S = 0$$
$$\vec{D} = \varepsilon \vec{E} = 0$$
$$div\,\vec{D} = \rho = 0$$

$$\varepsilon_2 \frac{\partial v_2}{\partial n} = -\rho_S$$

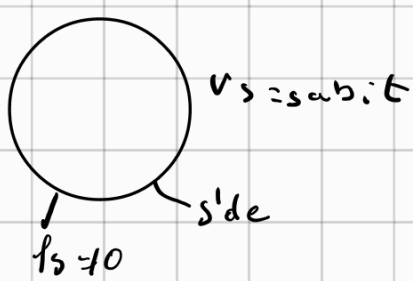
$$v_1|_S = v_S|_S = v_S = sabit$$

$$v_1|_S = v_S|_S = v_S = sabit$$
$$v_2|_{S=-\infty} < \infty$$

yüzey üzerinde potansiyel sabitse ve içeride yük yoksa iç bölgede potansiyel sabittir, elektrik alan da sabittir, ve içerde $\rho = 0$ //



$$\text{div}\,\vec{D} = \rho \Rightarrow \Delta V_1 = \frac{\partial^2 V_1}{\partial x^2} + \frac{\partial^2 V_1}{\partial x^2} + \frac{\partial^2 V_1}{\partial z^2} = 0$$



$$\frac{d^2 V_1}{dx^2} = 0 \Rightarrow V_1 = Ax + B$$

$$V_1|_{x=0} = A \cdot 0 + B = V_0 \Rightarrow B = V_0$$

$$V_1|_{x=d} = Ad + V_0 = V_d$$

$$\rho_{s1} = \vec{e}_x \left(\vec{D}_2|_{x=d}^{0} - \vec{D}_1|_{x=d} \right)$$

$$\rho_{s1} = -\vec{D}_1|_{x=d} \quad \vec{e}_x = -\varepsilon_1 \vec{E}_1 \quad | \vec{e}_x \quad x > d$$

$$\rho_s^{\,0} = -\rho_s^{\,d}$$